

<Header>

Group members: John Doe, Jane Smith

Bioinformatics

<Date>

## Tests

1. <our glorious protein 1> vs. <something else>
  - a. BLAST Alignment
    - i.
 

```
GRRPRRCWWRWPRCYWAPRATCTPERCVPAWISGTTSLGCMWRIA
G RPR+C   W      APR  C   R   A   S   SL       I+
GTRPRKC---W-----APRSSCRSSRNGCATRSIV-SL-----IS
```
    - ii. CountMatches: 16
    - iii. longestConsecutive: 3
  - b. Our Alignment
    - i.
 

```
GRRPRRCWWRWPRCYWAPRATCTPERCVPAWISGTTSLGCMWRIA
G RPR+C W  PR      R  +      C      T S++  +  I+
GTRPRKC-W-APRS--SCRSS--RNGCA-----TRSIV-SL--IS
```
    - ii. CountMatches: 13
    - iii. longestConsecutive: 3
2. <our glorious protein 2> vs. <another something else>
  - a. BLAST Alignment
    - i. etc.

## Discussion

Here we answer the questions posed on the assignment and give evidence based on our own specific tests. We consider the biological significance of our results in the context of the strengths and limitations of the Needleman-Wunsch algorithm, along with considering the inclusion of indels, the BLOSUM matrix, and various other factors. We also talk about how the biological significance may be improved at the algorithmic level. (This section may be as long as you want, but a lot of the points will come from how thoroughly you have analyzed your results, including what the results of your countMatches and longestConsecutive functions tell you about how the algorithm you implemented compares to BLAST.)

**Proteins used for testing: (including links so we can see online the real proteins you used. Please list all proteins in FASTA format.)**

- EMSY protein [Homo sapiens]

- >gi|19923559|ref|NP\_064578.2| EMSY protein [Homo sapiens]

MPVVWPTLLDLRDECKRILRKLELEAYAGVISALRAQGDLTKEKKDLLGELSKVLSISTER  
HRAEVRRRAVNDERLTTIAHNMSGPNSSSEWSIEGRRLVPLMPRLVPQTAFTVTANAVANA  
AIQHNASLPVPAETGSKEVVCYSYTTSTPTSTPVPSGSIATVKSPRPASPNVVLPSTVY  
VKSVCSDDEKPRKRRRTNSSSSPVVLKEVPKAVVPVSKTITVPVSGSPKMSNIMQSIANS  
LPPHMSPVKITFTKPSTQTTNTTTQKVIVTTSPSSTFVPNILSKSHNYAAVTKLVPTS  
VIASTTQKPPVVITASQSSLVSNSSSGSSSTPSPIPNTVAVTAVVSSTPSVVMSTVAQGVST  
SAIKMASTRLPSPKSLVSAPTQILAQFPKQHQSPKQQLYQVQQQTQQQVAQPSPVSHQQPQ  
QSPLPPGIKPTIQIKQESGVKIITQQVQPSKILPKPVATLPTSSNSPIMVVSSNGAIMTTKL  
VTTPTGTQATYTRPTVSPSIGRMAATPGAATYVKTTSGSIITVVPKSLATLGKKIISNIVSG  
TTTKITTIPMTSKPNVIVVQKTTGKGTTIQGLPGKNVVTLLNAGGEKTIQTVPTGAKPAILT  
ATRPITKMIVTQPKGIGSTVQPAAKIIPKIVYGQQGKTQVLIKPKPVTFQATVVSEQTRQLV  
TETLQQASRVAEAGNSSIQEGKEEPQNYTDSSSSSTESSQSSQDSQPVVHVIAARRQDWSE  
HEIAMETSPTHYQDVSSSESQATSTIKALLELQQTTVKEKLESKPRQPTIDLSQMAVPIQMT  
QEKRHSPESPISIAVVESELVAEYITTERTDEGTEVAFPLLVSQRSQPPSQPQRTLLQHVAQ  
SQTATQTSVVVKSIPASSPGAITHIMQQALSSHTAFTKHSEELGTEEVEEMDTLDPQTGLFYR  
SALTQSQSAKQKKLSQPPLEQTQLQVKTLQCFQTKQKQTIHLQADQLQHKLPQMPQLSIRH  
QKLTPQQEQAAQPKPDVQHTQHPMVAKDRQLPTLMAQPPQTVVQVLAVKTTQQLPKLQQAP  
NQPKIYVQPQTPQSQMSLPASSEKQTASQVEQPIITQGSSVTKITFEGRQPPTVTKITGSS  
VPKLTSPVTSISPIQASEKTAVSDILKMSLMQAQIDTNVEHMIVDPKKALATSMLTGEAGSL  
PSTHMVVAGMANSTPQQQKCRESCSSPSTVGSSLTTRKIDPPAVPATGQFMRIQNVGQKKA  
EESPAEIIIQAIPQYAIPCHSSSNVVVEPSGELLENNFTSQQLDDEETAMEQDIDSSTEDGTE  
PSPSQSSAERS

Link to protein data:

<http://www.ncbi.nlm.nih.gov/protein/blah/blah/blah>

---

- Protein 2

>gi|939576|gb|AAB212446.21| blah blah [Homo sapiens]

etc.